In-Class Assignment L7-1 (Group of 3 or less): Maximum-Weight Matching

(25 minutes) Submit to LEB2

You are holding a party and you want a musician to perform, a chef to prepare food, and a cleaning service to help clean up after the party. Three companies provide each of these three services, but one company can only provide one service at a time (i.e., Company B cannot provide both the cleaners and the chef). You are deciding from which company you should purchase each service to minimize the cost of the party. Each company charges the amount shown in the table below (in hundreds of dollars).

(a) Construct a weighted complete bipartite graph that models this problem as the maximum-weight matching.

(b) Execute lines 4-5 in the Kuhn-Munkres algorithm by showing feasible labeling of the graph and the corresponding equality subgraph.

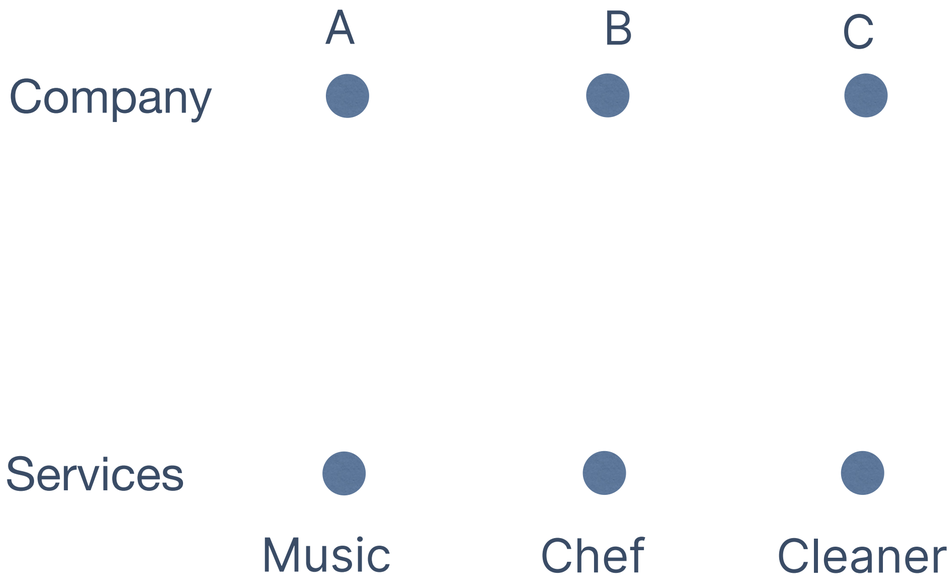
(c) Execute the while loop (lines 7-11) in the Kuhn-Munkres algorithm ***for one iteration*** by showing how the labels are improved and the equality subgraph is updated.

Company	Cost for Musician	Cost for Chef	Cost for Cleaner
Company A	11	12	15
Company B	15	13	18
Company C	12	15	25

Algorithm Hungarian (Kuhn-Munkres)

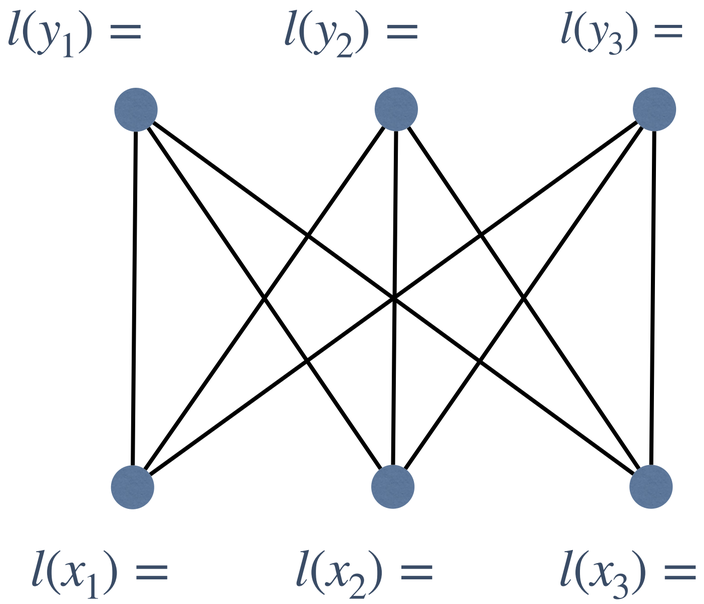
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Input: A weighted complete bipartite graph G = (A, B; E) where |A| = |B| =
Output: A maximum-weight perfect matching
Obtain from by using default vertex labeling
Obtain any arbitrary matching in
while is not perfect do
    Find an augmenting path and improve
    if Augmenting does not exist then
        Improve the labeling
        Update
return
```

(a) Weighted complete bipartite graph of the problem

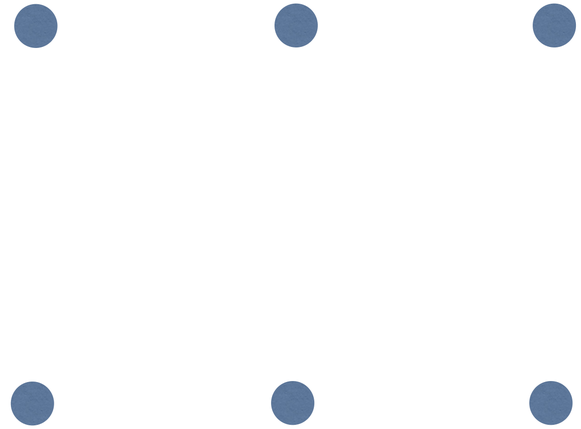


(b) Feasible Labeling and Equality Subgraph

G



G_l and Initial matching M



(c) BFS Forest on the Equality Subgraph

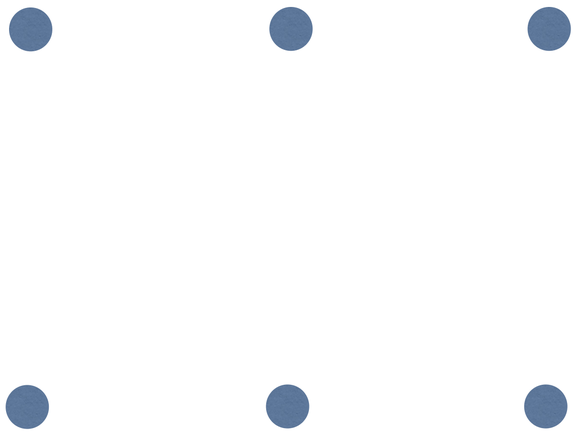
BFS forest

$F = (V_F, E_F),\ V_F = F_X \cup F_Y$

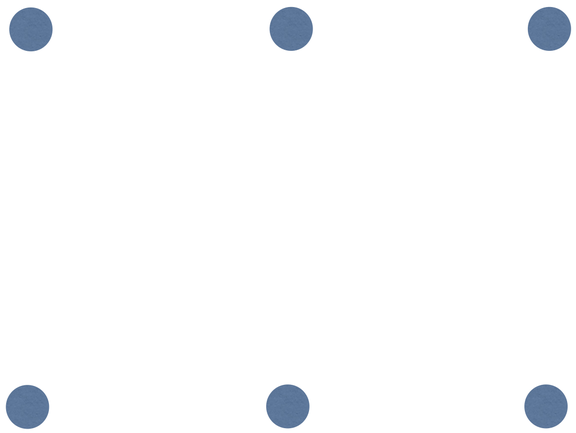
$F_X =$

$F_Y =$

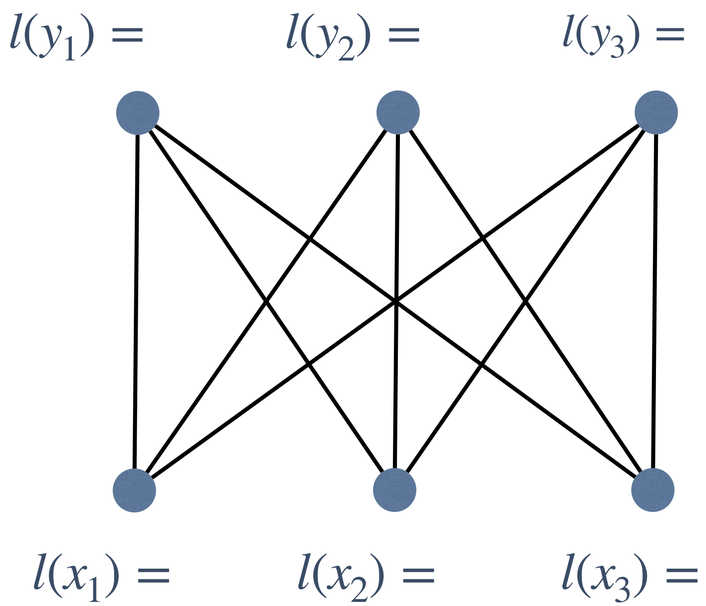
Equality subgraph G_l



Directed equality subgraph $G_{M,l}$



(c) Updated Labels



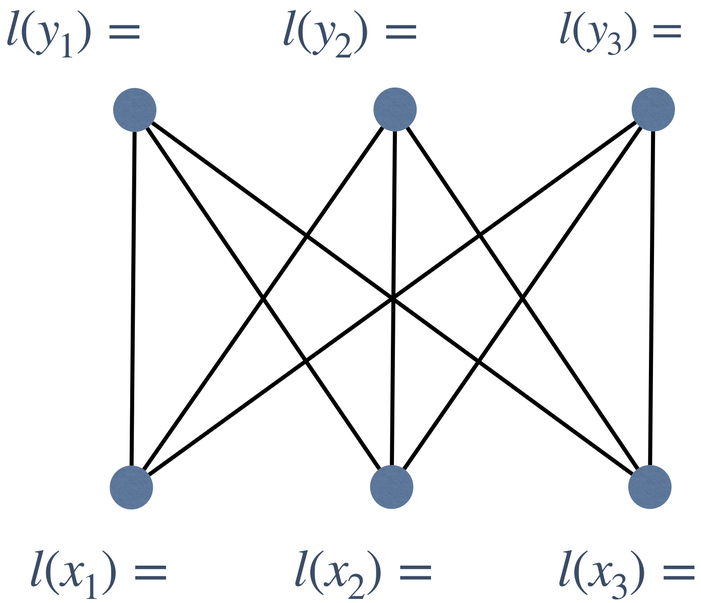
$F_X =$

$F_Y =$

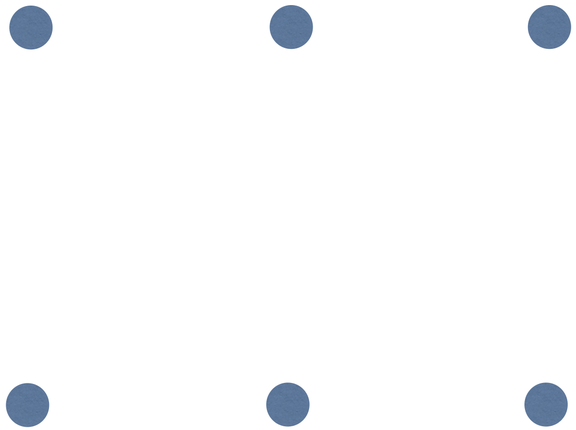
$= \min(\quad) + (\quad - \quad)$

$=$

(c) Updated Equality Subgraph

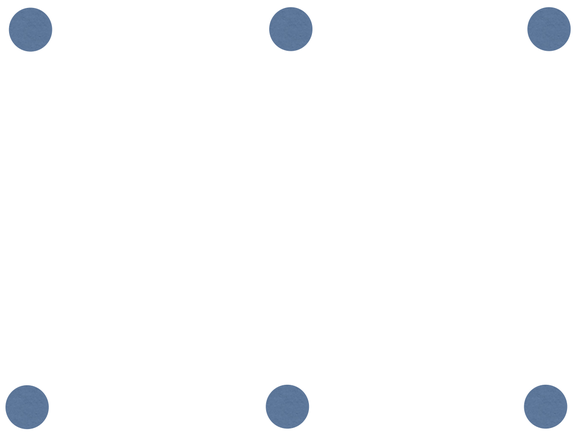


Updated equality subgraph G_l



(c) BFS Forest on the Updated Equality Subgraph

Updated directed equality subgraph $G_{M,l}$



BFS forest

$F = (V_F, E_F),\ V_F = F_X \cup F_Y$

$F_X =$

$F_Y =$